

STEADY

Between-Visit Care Command Center

Implementation-Ready Engineering Handoff

Clinical Intelligence Expansion | Document 03 | Repository-grounded against tprawlins-lang/Emdr | branch
claude/gifted-keller-501y5d | September 2026

Build directive

Turn Steady clinician home into a single between-visit action surface that combines existing safety/alert work with explainable non-safety clinical attention signals. It should tell the clinician who needs attention, why, what changed, what evidence supports it, and the one next action - without creating alert fatigue or pretending every deviation is urgent.

Shared product rule

The clinician decides what clinical meaning and action to accept. Steady may measure, organize, compare, and suggest, but it does not silently turn an association, model output, or behavioral signal into clinical fact.

Repository foundation this handoff assumes

Foundation	Current Steady anchor	Rule for this build
Longitudinal history	src/lib/spine.ts + ADR 0010	Durable state must be event-backed and replayable. Do not create a parallel mutable history.
Tenant/person boundary	src/lib/repository.ts + ADR 0011	Every new durable patient record carries tenant_id and person_id; new paths use TenantContext even while legacy paths are migrating.
Clinician patient workspace	src/app/clinician/member/[id]/*	New clinician surfaces extend PersonShell and existing evidence/review conventions.
Evidence-linked summary	src/lib/clinical/summary.ts	A generated claim without resolvable evidence is withheld. Summary is a reading aid, not the record.
Trajectory	src/lib/clinical/trajectory.ts	Separate scales stay separate; provenance remains visible; do not invent a composite clinical meaning from arbitrary axis alignment.
Work queue	src/lib/clinical/work-queue.ts	Clinician home opens on actionable work, stable server ordering, clear reason, one primary action.
Companion memory	src/lib/companion.ts	Patient/Companion data retains source class and memory exposure rules. AI output is not clinician evidence.
AI Gateway	ADR 0012	New model tasks must route through the gateway with schema validation, task version, provenance, evaluation, and PHI policy.

This series also assumes the Clinician Thoughts / Clinical Memory / Session Prep Engineering Specification v2.1 is implemented or being implemented as the common clinician-memory and evidence layer.

Cross-feature invariants

- **Safety authority stays deterministic.** These features may add clinician decision support but cannot clear, weaken, bypass, or replace the safety engine.
- **Stabilization scope stays intact.** No feature autonomously initiates trauma reprocessing, changes a gated treatment decision, diagnoses, or issues a treatment order.
- **Source classes never collapse.** Patient report, clinician observation, system measurement, formal note, Companion interaction, and model inference remain distinguishable.
- **Associations are associations.** A temporal or statistical relationship is not described as causation unless a clinician explicitly records that judgement from appropriate evidence.
- **No future-data leakage.** Historical reconstruction and point-in-time evaluations use only information Steady knew at the relevant cutoff.
- **Missing data stays visible.** Missing observations do not become zero, normal, compliant, or recovered.
- **Corrections append.** An authorized correction supersedes prior derived state without erasing history.
- **Model output is disposable; evidence and provenance are permanent.** Every material intelligence output points to evidence IDs and task/policy versions.

1. Product outcome

- Extend the existing work-queue philosophy rather than building a second dashboard.
- Safety alerts keep their existing authority and deadlines. New intelligence appears as non-safety attention signals with separate lifecycle and wording.
- Every row explains why it exists, what changed since prior review, newest evidence, owner, and one primary action.
- Handoff 04 and 05 will emit signals into this architecture. This handoff must therefore land before those engines.

2. Command Center information architecture

```
CLINICIAN TODAY
|
+-- Needs action now      existing safety / urgent obligations
+-- Review today          meaningful non-urgent change
+-- Follow up             explicit clinician follow-up / goal or recovery review
+-- Waiting on member     unanswered outreach / data gap stated as gap
+-- Waiting on staff      integration / referral / operational dependency
+-- Recently resolved
```

One row:

```
Person | reason | change | evidence time | owner | due | [one action]
|
+-- open detail drawer
    evidence
    source state
    why it appeared
    limits / what it does NOT mean
    action history
```

3. Durable attention signal model

Do not overload the existing alerts table with every non-safety intelligence output. Safety alert semantics must stay believable. Create a separate attention-signal projection that the work queue merges with alerts and caseload work.

```
CREATE TABLE clinical_attention_signals (
  id TEXT PRIMARY KEY,
  tenant_id TEXT NOT NULL,
  person_id TEXT NOT NULL,
  signal_type TEXT NOT NULL,
  source_feature TEXT NOT NULL,
  dedupe_key TEXT NOT NULL,
  attention_band TEXT NOT NULL CHECK (attention_band IN (
    'review_now', 'review_today', 'follow_up', 'watch'
  )),
  statement TEXT NOT NULL,
  change_text TEXT,
  state TEXT NOT NULL CHECK (state IN (
    'open', 'acknowledged', 'waiting_member', 'waiting_staff', 'resolved', 'dismissed'
  )),
  owner_person_id TEXT,
  due_at TEXT,
  first_detected_at TEXT NOT NULL,
  last_detected_at TEXT NOT NULL,
  evidence_at TEXT NOT NULL,
  policy_version TEXT NOT NULL,
  limitations_json TEXT NOT NULL DEFAULT '[]',
  created_at TEXT NOT NULL,
  updated_at TEXT NOT NULL,
  UNIQUE(tenant_id, person_id, dedupe_key)
);

CREATE TABLE clinical_attention_signal_evidence (
  signal_id TEXT NOT NULL,
  evidence_type TEXT NOT NULL,
  evidence_id TEXT NOT NULL,
  rank INTEGER NOT NULL,
  PRIMARY KEY(signal_id, evidence_type, evidence_id)
);

CREATE TABLE between_visit_care_actions (
  id TEXT PRIMARY KEY,
  tenant_id TEXT NOT NULL,
  person_id TEXT NOT NULL,
  clinician_person_id TEXT NOT NULL,
  signal_id TEXT,
  action_type TEXT NOT NULL,
  note TEXT,
  started_at TEXT,
  completed_at TEXT NOT NULL,
  duration_seconds INTEGER,
  outcome_state TEXT,
```

```
source_surface TEXT NOT NULL,  
created_at TEXT NOT NULL  
);
```

4. Signal provider interface

```
interface AttentionSignalProvider {  
  id: string;  
  version: string;  
  evaluate(args: {  
    tenantId: string;  
    personId: string;  
    evidenceCutoff: string;  
  }): Promise<AttentionSignalCandidate[]>;  
}  
  
Candidate includes:  
  type, dedupeKey, band, statement, changeText,  
  evidenceIds, evidenceAt, limitations, policyVersion  
  
Providers after this series:  
  existing-alert-adapter  
  explicit-followup-provider  
  return-to-life-provider  
  response-fingerprint-provider  
  recovery-trajectory-provider  
  therapeutic-load-provider
```

5. Initial providers in Handoff 03

Provider	Example signal	Guardrail
explicit follow-up	"You wanted to check sleep after the next visit."	Clinician-authored obligation; no AI priority invention.
Return-to-Life	"Grocery-shopping goal has new evidence and has not advanced across the current review window."	Use neutral "review" wording; no failure/compliance language.
Response Fingerprint	"Three recent sessions show delayed recovery burden after the same intervention."	Only when Handoff 02 support threshold is met; do not duplicate safety hard-stop alert.
Engagement gap	"No check-in for 9 days."	Fact and reason to ask, not predicted disengagement.
Existing caseload/alerts	Current safety and operational work	Safety bands/deadlines remain authoritative.

6. Ordering policy

- Existing immediate/high safety obligations remain first.
- Within non-safety signals: overdue explicit follow-up before review_today; then evidence recency and policy-defined clinical importance; stable ID final tiebreak.
- A model does not sort the queue. Ordering is deterministic server policy.
- Later trajectory/load signals cannot leapfrog a safety obligation by assigning themselves a higher score.
- Duplicate same-person same-reason signals collapse and preserve event/evidence count.

7. Signal lifecycle

```
candidate detected  
  |  
  v  
OPEN ---- clinician reviews ----> ACKNOWLEDGED  
  |                               |  
  |                               +--> RESOLVED  
  |                               +--> WAITING_MEMBER  
  |                               +--> WAITING_STAFF  
  |                               +--> DISMISSED (reason recorded)  
  |  
  +-- evidence changes --> UPDATE same dedupe_key, do not spam duplicate row
```

If resolved condition recurs with genuinely new evidence:
REOPEN existing lineage or create versioned recurrence according to signal policy

8. Clinician actions

Action type	Behavior
review	Open evidence/Session Prep and record reviewed state.
contact	Launch existing messaging/contact workflow; do not send automatically.
add_followup	Create clinician follow-up linked to evidence.
record_thought	Launch Record Thoughts in the patient context.
adjust_plan_link	Navigate to clinician-controlled plan surface; signal itself cannot alter plan.
resolve	Close work item with reason/outcome.

9. Care-time ledger

- Capture clinician between-visit work time where practical because it supports operations, staffing, and future reimbursement workflows.
- Do not mark time as billable solely from Steady data. Store action type, timestamps, duration, source signal, and outcome; billing eligibility remains a separate policy/integration layer.
- A clinician can correct or annotate care time. Do not use hidden background page time as clinical work duration.

10. UX

- **Clinician Today:** one work queue, group counts, freshness timestamp, no parallel "AI alerts" list.
- **Row:** person, plain-language reason, "new since your last review" when true, evidence timestamp, owner, due state, one action.
- **Detail drawer:** evidence chain, why this appears, limitations, source feature/policy version, previous actions, related Session Prep.
- **Patient overview:** "Active work" continues to use the same projection filtered to the patient.

11. AI role

- No AI task is required to decide whether a work item exists or where it sorts.
- Optional AI can compress evidence into a human-readable explanation only after a deterministic provider emits the candidate. The original provider statement remains available.
- AI cannot set due date, owner, urgency, safety state, or resolve an item.

12. Events

Event	Meaning
attention_signal.opened.v1	New durable non-safety work item.
attention_signal.updated.v1	Same dedupe lineage changed due to new evidence.
attention_signal.acknowledged.v1	Clinician reviewed without resolving.
attention_signal.state_changed.v1	Waiting/resolved/dismissed lifecycle.
between_visit_care.action_recorded.v1	Clinician action/time/outcome recorded.
attention_signal.reopened.v1	Resolved condition recurred with new evidence under policy.

13. Implementation phases

Phase 1 - Signal substrate

Build

- Attention signal schema + evidence
- Provider interface
- Merge into work queue projection

Definition of done

- Existing queue behavior unchanged when feature off
- Stable dedupe/order tested

Phase 2 - Initial providers

Build

- Follow-ups
- Return-to-Life
- Response Fingerprint
- engagement gap adapter

Definition of done

- No duplicate safety alert semantics
- Every row has reason/evidence/action

Phase 3 - Action ledger + UX

Build

- Detail drawer
- action commands
- care-time ledger

Definition of done

- Action history auditable
- No passive browser time counted

Phase 4 - Downstream extension contract

Build

- Document provider API for Trajectory and Load
- Feature flags and policy registry

Definition of done

- Handoffs 04/05 can add providers without editing queue semantics

14. Acceptance criteria

- Clinician has one unified queue, not an alerts list plus an AI list plus a caseload list.

- Safety work remains visibly and semantically different from non-safety review signals.
- Signal generation and queue order are deterministic and versioned.
- Repeated evidence updates one row rather than flooding the queue.
- A dismissed/reviewed signal has an auditable human action.
- Later feature providers can plug in without changing the queue contract.

Appendix A. File-level implementation map

Path	Change
src/lib/clinical/attention-signals.ts	NEW signal lifecycle/repository.
src/lib/clinical/work-queue.ts	EXTEND to merge attention signals while preserving stable server ordering.
src/lib/clinical/attention-providers/*	NEW provider registry.
src/app/clinician/today/page.tsx	Update unified queue rendering if needed.
src/components/clinical/WorkQueueRow.tsx	Add source/limitations/detail support without multiple row actions.
src/components/clinical/AttentionSignalDrawer.tsx	NEW evidence/action drawer.
src/lib/spine.ts + projections.ts	Add signal/action events.
src/lib/db.ts + scripts/pg-schema.sql	Add/mirror schema.
tests/attention-signals*.test.ts	Dedupe, order, action lifecycle, tenant security, replay.